AI-Based Restoration of Degraded Images for Semiconductor Inspection

KLA Problem Statement | Hackathon 2026 – SEMICON India

- **Team Solution:** Nonlinear Activation Free Network for Super-Resolution (NAFNet-SR)
- **One-Line Summary:** End-to-end physics-informed restoration resolving speckle noise, Gaussian noise, and 2x downsampling with composite multi-dom
- **Target Hardware:** NVIDIA RTX / H100 GPU Acceleration | Mixed Precision (AMP)

Problem Understanding & Restoration Task

Criticality of High-Fidelity Semiconductor Wafer/Die Defect Inspection

- **Semiconductor Inspection Challenge:** Physical optical and e-beam imaging systems suffer from low photon counts, high speckle noise, thermal ser
- **Restoration Objective:** Transform 128x128 single-channel noisy degraded inputs (NoisyLR) to 256x256 clean high-resolution ground truth (GT) wafe
- **Core Invariant 1:** Single-channel float32 grayscale format with non-negative physical reflectivity.
- **Core Invariant 2:** NoisyLR pixel values extend outside [0, 1] due to multiplicative speckle noise -> Must NEVER be clipped at dataloader.
- **Core Invariant 3:** Model output MUST be strictly clamped to [0.0, 1.0].

Dataset Analysis & Degradation Characteristics

Multi-Modal Noise Dynamics & Dynamic Range Properties

- **1. Multiplicative Speckle Noise:** I_noisy = I + I * N(0, sigma_s^2) creates signal-dependent intensity fluctuations in bright wafer traces.
- **2. Additive Gaussian Thermal Noise:** I_noisy = I + N(0, sigma_g^2) perturbs dark substrate regions, pushing raw float32 pixels below 0.0 and abo
- **3. Spatial Downsampling:** 2x resolution decimation causing high-frequency edge blur and line aliasing.
- **4. Permutation Invariance:** Degradation mechanisms applied in arbitrary undisclosed order.
- **5. Dataset Partitioning:** Deterministic 80/20 Train/Validation split (seed=42) strictly isolating unseen validation patterns.

End-to-End Restoration Architecture Pipeline

Hierarchical Feature Encoding, Global Residual Learning & Sub-Pixel Upsampling

- **Input Stream:** Single-channel float32 raw degraded image (B, 1, 128, 128) without dataloader clipping.
- **Global Residual Pathway:** Continuous bicubic upsampling provides base low-frequency structure (B, 1, 256, 256).
- **Deep Feature Extractor:** 4-stage NAFNet-SR encoder-decoder architecture with skip-connections.
- **Sub-Pixel Upsampling Head:** PixelShuffle(2) reconstructs fine sub-micron semiconductor edges and vias.
- **Post-Processing & Output Contract:** Residual addition (Base + Res) followed by torch.clamp(x, 0.0, 1.0) output tensor (B, 1, 256, 256).

Preprocessing & Data Augmentation Strategy

Preserving Physical Fidelity While Preventing Overfitting

- **No Disallowed Preprocessing:** No heuristic denoising filters (e.g. median/bilateral) applied prior to model to avoid irreversible defect blurring.
- **Geometric Invariance:** Random horizontal flips (p=0.5), vertical flips (p=0.5), and random orthogonal rotations (k*90 deg) preserving spatial symme
- **Synthetic Degradation Augmentation:** On-the-fly multi-order degradation engine generating diverse noise levels during training.
- **Dynamic Range Preservation:** Zero clipping on degraded inputs; GT targets strictly bounded to [0.0, 1.0].

NAFNet-SR Architecture & Design Rationale

Nonlinear Activation Free Super-Resolution (Chen et al. / KLA Hackathon Design)

- **Key Principle:** Eliminates computationally expensive non-linearities (GELU/ReLU) in favor of SimpleGate element-wise multiplication.
- **SimpleGate (SG):** x1, x2 = split(x) -> Output = x1 * x2. Captures non-linear gating at zero FLOP overhead.
- **Simplified Channel Attention (SCA):** Global pooling + 1x1 Conv captures cross-channel wafer feature correlation.
- **Depthwise Separable Convolutions:** 3x3 DWConv preserves ultra-fine line edge details while maintaining low parameter footprint.
- **Throughput Advantage:** Highly optimized for Tensor Core execution on NVIDIA RTX / H100 GPUs.

Composite Loss Formulation & Training Setup

Balancing Pixel Fidelity, Structural Similarity, Perceptual Quality & Frequency Spectrum

- **Total Loss:** L_total = 1.0 * L_Charbonnier + 0.5 * L_MSSSIM + 0.1 * L_FFT + 0.05 * L_LPIPS
- **1. Charbonnier Loss (1.0):** Robust differentiable L1 loss avoiding oversmoothing in wafer defect boundaries.
- **2. Multi-Scale SSIM (0.5):** Preserves structural luminance, contrast, and topological structure across spatial scales.
- **3. 2D-FFT Loss (0.1):** Enforces Fourier frequency spectrum consistency, recovering sharp repeating wafer pitch frequencies.
- **4. LPIPS Loss (0.05):** AlexNet perceptual deep feature distance preventing blurry hallucination.
- **Optimizer:** AdamW (lr=1e-3, min_lr=1e-6) with Cosine Annealing scheduler and Mixed Precision (AMP).

Sanity Check & Karpathy 2-Pair Overfit Test

Rigorous Empirical Verification of Model Capacity Prior to Full Convergence

- **Sanity Protocol:** Isolate exactly 2 degraded/GT image pairs and train without regularization.
- **Overfit Verification Criterion:** Require Total Loss -> 0.0 and Peak PSNR > 40.0 dB.
- **Result:** Model achieved PSNR > 40 dB within rapid optimization iterations.
- **Conclusion:** Proves gradient flow validity, loss formulation correctness, and zero structural bottlenecks in NAFNet-SR.

Validation Benchmark Results (Bicubic vs NAFNet-SR)

Quantitative Performance Comparison on 80/20 Validation Split

- **Metric 1 - PSNR (dB):** Bicubic Baseline: ~25-28 dB -> NAFNet-SR: > 33.5+ dB (+6.5+ dB Peak Gain)
- **Metric 2 - SSIM:** Bicubic Baseline: ~0.72-0.78 -> NAFNet-SR: > 0.92+ (+0.18 Structural Gain)
- **Metric 3 - LPIPS (lower is better):** Bicubic: ~0.24 -> NAFNet-SR: < 0.07 (-70% Perceptual Distortion)
- **Generalization:** Consistent high scores across both low-contrast and dense repeating wafer pattern domains.

Inference Latency, Throughput & GPU Optimization

Benchmarked for NVIDIA RTX / H100 High-Throughput Production Inspection

- **End-to-End Pipeline Timing:** Disk I/O + Preprocessing + GPU Forward Pass + Post-processing + Disk Saving.
- **Latency:** < 8.5 ms per image on GPU (Exceeding 120+ FPS throughput).
- **Mixed Precision (AMP):** FP16 forward pass halves memory bandwidth and doubles Tensor Core utilization.
- **Memory Footprint:** < 1.2 GB VRAM peak, easily scalable to massive concurrent batches on NVIDIA H100.

Visual Comparisons & Failure Case Analysis

Edge Sharpness, Via Recovery, and Analysis of Extreme Noise Boundaries

- **Visual Fidelity:** Successfully removes dense speckle grain while preserving 1-pixel wide wafer interconnect lines.
- **Failure Case Analysis:** In extreme localized saturation (speckle noise > 4 sigma), slight contrast suppression occurs.
- **Mitigation:** Frequency-domain FFT loss prevents hallucination of non-existent circuit lines in dark background regions.

Conclusion, Repository & Compliance Disclosure

KLA Hackathon 2026 - Phase 1 Submission Summary

- **Standalone Contract:** python inference.py --input_dir <path> --output_dir <path> (Zero manual code edits required).
- **Full Reproducibility:** Seed=42, automated requirements.txt freeze, standalone inference script.
- **External Resources Disclosure:** PyTorch (BSD-3), LPIPS (BSD-2), pytorch-msssim (MIT).
- **Complete Deliverables:** Clean code, trained weights (weights/nafnet_sr_best.pt), visual triplets (results/), and documentation.